

SOP-STRUCT-3400: Structural Steel and Foundation Inspection

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Section 1. Purpose and Scope

Governs periodic inspection of load-bearing structural steel, pipe racks, and equipment foundations for corrosion, deformation, and settlement.

Section 2. Inspection Interval

Clause 2.1 — General structural inspection is conducted every 5 years; structures in a corrosive or coastal environment are inspected every 3 years.

Clause 2.2 — A structure exposed to an abnormal load event (vehicle impact, seismic event exceeding design basis, fire) requires inspection within 30 days of the event, regardless of the routine schedule.

Section 3. Findings Requiring Engineering Evaluation

Clause 3.1 — Section loss (corrosion) exceeding 10% of a member's original cross-section, visible weld cracking, or foundation settlement exceeding 1/2 inch differential requires engineering evaluation before the structure is certified fit for continued service.

Section 4. Temporary Support

Clause 4.1 — A structural finding presenting immediate collapse risk requires immediate area exclusion and temporary shoring/support before any further engineering evaluation proceeds; the area shall not remain in normal use pending evaluation.

Section 5. Records

Clause 5.1 — Inspection reports and engineering evaluations are filed to the Civil/Structural Integrity File and retained for the operating life of the structure.